A Systematic Review of Evolving Ransomware Techniques and Defense Mechanisms in Critical Infrastructure Sectors Post-2017

IT22562074 | JAYARATHNA K.P.G.C.M | Mobile – 070-1463982| Y3.S2.WE.CS.01.02

Abstract— Attacks involving ransomware have developed rapidly over the past few years and pose a serious risk to the critical infrastructure of an organization, such as healthcare, energy, and transportation services. Cybercriminals have adopted new tactics post 2017 including double and triple extortion schemes, Ransomware as a Service (RaaS), and more targeted attacks on critical services. This study review synthesizes literature conducted after 2017 focusing on emerging ransomware methods and their corresponding defense strategies. Using PRISMA methodology to filter relevant literature, we analyzed 25 peer-reviewed articles. Key findings illustrate increased sophisticated ransomware attacks exploiting social engineering, supply chain compromises, and zero-day vulnerabilities. Proactive defense mechanisms now emphasize using anomaly detection systems, Zero Trust models, and threat intelligence frameworks. The previously mentioned gaps remain conspicuously present and suggest further threats that require immediate attention. This paper provides an up-to-date overview of the threats and vulnerabilities facing critical infrastructure, disclosing primary gaps in research that need addressing and strengthen resilience concerning critical infrastructure.

Keywords— Ransomware, Triple Extortion, Double Extortion, Critical Infrastructure Protection
Cybersecurity, Ransomware-as-a-Service (RaaS), Defense Mechanisms, Post-2017 Threat Landscape, Anomaly Detection Systems

Introduction

In the past 10 years, ransomware has become one of the most financially damaging forms of cybercrime. It is classified as malicious software that encrypts data and demands a ransom for the decryption keys. Ransomware now targets governments, enterprises, and critical infrastructure and has shifted from opportunistic attacks on individual users. This evolution has included more advanced monetization methods, business models, and technical sophistication of attacks, one being Ransomware-as-a-Service (RaaS).
Healthcare, energy, transportation, water supply, and public services constitute essential infrastructure and have become more susceptible due to interconnected automated systems, legacy software, and potential for catastrophic disruption. The 2021 Colonial Pipeline attack and the 2021 Irish Health Service Executive attack demonstrate the concerning spread of ransomware attacks on critical infrastructure that can have broad economic or life-critical impact. The security environment has changed fundamentally since the year 2017. Recent modern ransomware versions apply both double extortion, which is stealing the data first and after encrypting it, and triple extortion—pressuring victims by offering threats to the clients or making public disclosures. These changes have transformed ransomware from merely a threat to the availability of data into multi-layered threat to organizational and national security.
The purpose of this review is to investigate the progress in techniques of ransomware and its associated defensive strategies in critical infrastructure sectors after 2017. Following the PRISMA methodology, this paper analyzes 25 peer-reviewed articles with the purpose of identifying emerging trends, assessing current defensive strategies, and exposing the existing gaps in literature. These results will help explain the adaptations of techniques used in ransomware and outline how essential services must be defended concerning future threats.

Methodology

This review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure a transparent and reproducible selection process of academic literature related to ransomware threats and defense mechanisms in critical infrastructure sectors post-2017.

Data Sources

A comprehensive literature search was conducted using the following academic databases:

- IEEE Xplore
- Scopus
- ScienceDirect
- SpringerLink
- ACM Digital Library
- Google Scholar (for grey literature and supplementary sources)

The search was restricted to peer-reviewed journal articles, conference proceedings, and white papers published between January 2017 and March 2025.

Criteria	Inclusion	Exclusion
Publication Date	2017–2024	Before 2017
Relevance	Focus on ransomware in critical infrastructure	General malware or unrelated cybersecurity threats
Document Type	Peer-reviewed articles, conference papers, white papers	Editorials, newsletters, blogs
Language	English	Non-English
Sectors	Healthcare, energy, water, finance, transport	Consumer tech, education (unless critical sector related)

Table 01: Inclusion and Exclusion Criteria

D. Selection Process

After initial retrieval of 106 papers, duplicates were removed, and abstracts were screened for relevance. A full-text analysis of 38 papers was performed, resulting in a final selection of 25 high-quality publications used for this review.

I. Ransomware Evolution After 2017

Ransomware attacks from 2017 onwards have been characterized by greater sophistication in attack vectors, monetization models, and overall strategy. The more straightforward ransomware attacks involving encryption and payment demands in Bitcoin have made way for more complex and aggressive strategies that are carefully planned.

A. Pre-2017 Method : Traditional ramsomeware

Ransomware campaigns like CryptoLocker (2013) and Locky (2016) used indiscriminate email spam and basic encryption routines as their primary methods before 2017. Basic encryption was their go-to method and spam email were their favorite form of market correspondence. Moreover, these shoots were used to target individuals and small organizations which utilized social engineering techniques as well as unpatched software vulnerabilities.

B. Notpetya & WannaCry: 2017 Change in Paradigm

The extortionate attacks by CryptoLocker and Locky were given a tremendous paradigm shift with the emergence of WannaCry in May 2017. In WannaCry, Microsoft's SMB protocol EternalBlue exploit (CVE-2017-0144) was employed for infecting over 230,000 systems in over 150 countries.This latter mark has gone on to be used as a criterion of measurement of its infection statue. The impact on UK's NHS ransomware highlighted critical infrastructure as an impotent target. [1]

NotPetya attempted to disguise itself as ransomware in June of 2017, but it was a wiper that attacked Ukrainian infrastructure with geopolitical motives disguised as weapons. Restraints were posed on how technology could be misused. [2]

C. Post 2017 Improvements: Targeted attacks with ransom demands

Today's ransomware is more sophisticated as they have transitioned from a volumetric approach to focusing on precision and depth. There has been an increase of attention provided to high-level strategic features by Ryuk, Maze, REvil, Conti and LockBit towards critical infrastructure when conducting reconnaissance for weeks prior to releasing Trojan viral payloads.

A landmark development in this area is the introduction of double extortion pioneered by the Maze group, where encryption of the data is done post exfiltration and public release is threatened if the ransom demand isn't fulfilled with adequate funds regardless of whether the funds are met. [3] This caused a ripple effect of triple extortion tactics where campaigners Avaddon and SunCrypt used partners of the targeted businesses as additional weapons to apply pressure alongside DDoS attack alongside data stealing, encryption and blackmailing. [4]

D. Emergence of Ransomware-as-a-Service (RaaS)

With the RaaS model, ransomware became easier to exploit as unscrupulous business affiliates were able to hack and profit by using sophisticated ransomware kits developed by others. Constructs like DarkSide, Conti, and LockBit operate on revenue-sharing models, which have myriad vendors and lower market gatekeeping. [5]

E. Sector-Specific Targeting

After 2019, the focus shifted to Critical Infrastructure. The Colonial Pipeline attack (2021) caused DarkSide to cripple fuel supply chainss into the US Eastern Coast. Also in 2021, Conti attacked the Irish Health Service Executive (HSE), bringing healthcare systems to a complete standstill, demonstrating the impact of ransomware on critical functions in society. [6]

F. Technical Sophistication

Modern-day ransomware comes with features like modular design, methods for circumventing sandboxes, anti-debugging protective measures, and lateral network movement using legitimate resources such as Cobalt Strike, Mimikatz, and PowerShell. Encryption techniques have also evolved to hybrid systems (AES plus RSA), making it almost impossible to decrypt without the attacker's private key.

FIGURE 1: History of Ransomware

II. . Advanced Techniques: Double & Triple Bartering

The methods employed in ransomware attempts have changed remarkably since 2019. The expansion of cybercriminal groups embracing double and triple extortion strategies have created more stress for victims. These methods now include not just encryption, but also data leaking, exposure, and third-party intervention to boost the success levels for ransom while maximizing profit simultaneously. [7]

A. Double Extortion: Data Theft + Encryption

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, sc, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

The double extortion approach gained notoriety with the Maze ransomware group in 2019. Unlike common ransomware that encrypted files, Maze actors first accessed sensitive data and stole it with the threat of posting it online unless a payment was made. This advance threat model greatly improved the chances of ransom being paid, especially from firms dealing with confidential or regulated information such as PII, PHI, and financial data".
Other ransomware families followed in their footsteps, like REvil (Sodinokibi), DoppelPaymer, and Conti. Attackers frequently leak some of the datasets on "leak sites" hosted on the dark web, named "Happy Blog" by REvil or Conti named "Conti News" site, to force the victims into conceding to their demands and prove they have the information they claim to possess. [8]

1) Revil & JBS Foods Case Study (2021)

REvil attacked JBS Foods in June 2021. JBS Foods, one of the largest meat suppliers globally, was struck in an attempt to cripple food production and food logistics services in the United States and Australia. JBS Foods eventually paid $11 million in ransom to avoid sensitive data being leaked and to have systems back online. This demonstrated how double-extortion schemes can rapidly affect critical infrastructure. [9]

B. Third-Party & DDoS: Triple Extortion

The addition of external actors to apply pressure in the campaign enhances the double-extraction model. This may involve:

- Persuading the media and regulators to publish the data
- Attacking business partnered customers or suppliers
- Conducting DDoS attacks, which hinders operational capabilities

Avaddon, Ragnar Locker, and SunCrypt are known to adopt this model. These groups frequently reach out to clients of the victim organization in order to maximize risk to reputation and legal repercussion exposure. [10]
Case Study: Vastaamo Psychotherapy Center (Finland 2020)
A breach was committed on Finnish psychotherapy firm, Vastaamo, where the attacker did not just threaten the organization but also targeted individual patients. In what is recognized as one of the first public examples of a healthcare provider becoming a victim of triple extortion tactics, sensitive therapy records were leveraged in a multi-layer extortion scheme. [11]

C. Impacts: Financial, Legal, and Reputational

The following categories summarize the combined effects that advanced extortion models inflict on targeted organizations:

- Financial: Downtime, recovery, legal action, and post-financial ransom, which costs several million dollars, coupled with the fines incurred from data protection legislation such as GDPR or HIPAA, also add up to substantial expenditures.

- Legal: Data leaks necessitate breach disclosure, within a particular jurisdiction, the payment of ransom may also be subject to prohibitive anti-terrorism or anti-money laundering legislation.

- Reputational: Strategic public trust is lost from customers, business partners, and data stakeholders as a result of public data leaks. This impacts healthcare, finance, and utilities more severely since they serve the public directly.

Security experts suggest such aggressive tactics elevate the severity of considering ransomware an IT issue to that of an organizational problem, thus needing an integrated legal, communication, and cybersecurity strategy to address it. [12]

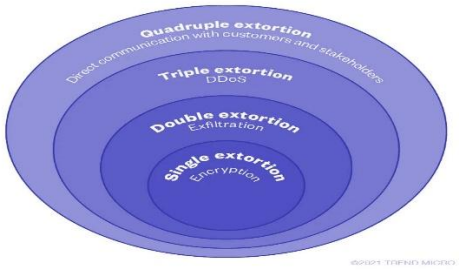

FIGURE 2: Phases of Ransomware Extortion

III. Ransomware-as-a-Service (RaaS)

By far the most innovative and popular approach of distributing Ransomware for unaffiliated cybercriminals is through Ransomware-as-a-Service (RaaS). The evolution of RaaS has given rise to numerous attacks run by

technologically deficient individuals, often referred to as "script kiddies.") RaaS occurs when ransomware is sold as an easy-to-use product. Automated attack kits come with instructions, access to customer support, and entire business models to buy into. [13]

A. *The Business RaaS Model*

Ransomware service models are akin to other software rental opportunities. For instance, developers create ransomware payloads which they then sell, and affiliates are tasked with marketing them, infecting victims' systems, and collecting ransom payments. [14] Developers also get paid 20-30% of total ransom payout for their work, which is a payment model used in more modern software-as-a-service (SaaS) available today. [15]Paying through these dollars comes with many benefits such as:

- Profit instruments where each user gets assigned a unique value depending on the infections caused and payment made
- A user interface for not just viewing but also tracking other things like payments
- Payment made to technical assistance or customization resources provided on call
- A forum where users are supported all the way from vetting to recruiting

Groups such as REvil, Conti, DarkSide, LockBit and BlackCat (ALPHV) adopt the RaaS model and change their tactics based on market shifts and affiliate behavior. [16]

B. *Dark Web Actors and Their Markets*

Ransomware RaaS teams work on Exploit, RAMP, and XSS darknet forums using encrypted communication channels and trust-based escrow services. Phishing ads are often hired based on geographic targeting, which is a skill of enticing pseudo identities. [17]

According to reports, the average ransom request in RaaS attacks skyrocketed from $110,000 in 2019 to over $800,000 in 2023. The best affiliates are said to earn six-figure payouts for each attack. Some groups have become RaaS agile, with the temporary disbanding and rebranding (e.g. REvil's disappearing act then returning, Darkside turning into BlackMatter) to escape law enforcement shakedowns. [18]

An increase of lower sophistication cyber-attacks, in particular targeting underfunded critical infrastructure entities with minimal to no access mature defensive security controls, has been noted.

IV. Impacts by Sector on Critical Infrastructure

Ransomware attacks have been rising to target key infrastructure sectors due to their intricate operations, confidential information, and limited cybersecurity infrastructure.The collective impact includes economic retardation, cyber interruptions, public health hazards, and security concerns.

A. *Healthcare*

Due to the sensitive nature of patient data and low tolerance for downtime, the healthcare field is considered to be one of the most impacted areas. One of the worst attacks occurred in 2020 against Universal Health Services, which is one of the biggest hospital systems in the U.S., after the complete shutdown of over 400 systems which forced manual record keeping. [12]

The Conti ransomware subgroup hacked into Ireland's Health Service Executive during May 2021. This attack effectively paralyzed a large proportion of care delivery and diagnostic IT infrastructure at a national scale. Recovery costs were projected to estimated €100 million.

B. *The Energy Industry*

The DarkSide Group attack on the Colonial Pipeline in May 2021 was a major turning point.Their ransomeware attack encrypted crucial operational systems which resulted in the pipeline shutting down that caused 45% of fuel supply to the east cost of the United States to be cut. Even after paying a ransom of $4.4 million, the attack forced the federal government to decare a state of emergency while other changes to policy were made, such as mandatory cybersecurity incident reporting for pipelines.

C. *Transportation, Water, and Manufacturing*

Transportation and logistics have not escaped the reach of ransomware attacks either. Transnet, a logistics company, in South Africa in 2021 was faced with a ransomware attack that suspended port operations and delayed shipments globally. Hence, alarm bells clearly seemed to have rung because of the new ransomware attacks on the water utilities in Florida and Israel, showing plans are underway to take water supply systems hostage.

In the year 2019, LockerGoga ransomware was a bane to Norsk Hydro and crippled its production in several countries, making the company suffer losses in excess of $60 million. Such attacks are a testament to the fact that ransomware can disrupt not only the functioning of digitally controlled frameworks, but also systems that rely on industrial machinery. [14]

V. Defense Mechanisms

Ransomware attacks and their counter-defense strategies have developed alongside each other throughout history. After 2017, algorithms protecting important infrastructure moved from reactionary-response techniques to multi-layered, proactive like behavior-based detection. AI vigilant anomaly monitoring along with applying zero trust and MITRE ATT&CK frameworks also emerged to up guard infrastructure against ransomware attacks. [19]

A. *Signature-Based vs. Behavior-Based Detection*

The signature-based detection strategy relies on known malware fingerprints signature like hashers, file names, and byte sequences. [20]This method is quick and easy to

implement but does not work against polymorphic, fileless, zero-day, or variant assaults. [21]
Behavioral analytics that track user activity also known as behavior-based detection do flags suspicious activity like:

- Abnormal file sharing
- Power Shell or WMI script oddities
- Shadow copy creations or task scheduling

Novel strains of ransomware have a hard time getting through behavioral models. Microsoft Defender ATP, Crowdstrike Falcon and Sophos Intercept X use behavioral analytics to track and intercept ransomware in real-time.

B. Anomaly Detection Systems

Employing statistics, ML, and heuristics, anomaly detection systems focus on identifying changes in behavior on a particular endpoint or network. Anomaly detection performs really well on critical infrastructure domains due to their deterministic workflows, for instance in SCADA systems used in power and energy. [22]

Examples:

- In PLCs (programmable logic controllers), unusual traffic has been noted.
- CPU as well as disk I/O activity show unexpected spikes.
- Communication deemed abnormal with external IPs is observed.

Darktrace, Nozomi Networks, and Cisco Stealthwatch instance learn to detect anomalies and use heuristics in their lower-level industrial control networks.

C. Defenses Based on AI and ML

The ongoing issues with ransomware have been addressed AI and ML technologies. Techniques applying IoT signature methods are slower in detection logic as compared to identification methods grounded on detection patterns. AI/ML-based solutions train models with IV telemetry datasets in order to recognize mediators.

Supervised learning approaches, e.g. Random Forest and SVM, classify identified threats while other, unsupervised ones, like K-Means and autoencoders, focus on identifying new, unheard-of activities. Reinforcement learning is under study for the purpose of creating reactive automated defense systems that change according to an attacker's actions during an ongoing cyber-assault.

Nonetheless, there are:

- Sensitive context environments with prominent false positive rates.
- Adverse shifts in model performance over time within changing network conditions.
- Attempts to sabotage detection systems through adversarial ML, or outsmart them entirely sottod satellites while evasion mechanics are in place.

D. Zero Trust Architecture (ZTA)

Architecture where the idea is “never trust, always verify.” Focuses on continous authentication and a combination of segmentation and least-privilege access. [22]

Main characteristics include:

- Micro segmentation of network regions
- Multi-factor authentication (MFA)
- Role-based access controls (RBAC)
- Identity verification alongside device posture assessment

ZTA, or Zero Trust Architecture, mitigates vertical as well as lateral movement post initial breach in critical infrastructures. ZTA frameworks are published by CISA, NIST, and Microsoft, and are gradually gaining traction in the energy and healthcare industries.

E. Incident Response Frameworks (NIST & MITRE ATT&CK)

Effective response (IR) capabilities are critical to ransomware containment and recovery.

The NIST Cybersecurity Framework (CSF) is comprised of 5 functions: Identify, Protect, Respond, and Recover, which are utilized by many operators of critical infrastructure. [23]

Conversely, MITRE ATT&CK frameworks inventory adversarial tactics, techniques and procedures (TTPs). Red teams use ATT&CK to conduct simulations of ransomware campaigns to examine adversarial behavior (e.g., lateral movement via RDP, data exfiltration via DNS), building visibility in the SOC.

ATT&CK MITRE also assists in threat hunting by mapping real world campaigns such as Conti or LockBit to their kill chain techniques (e.g, T1059: Command and Scripting Interpreter).

VI. Closing the Gaps in Research

Although numerous strides have been made since 2017 to improve ransomware detection and mitigation, both academically and practically there are still significant gaps, especially in the area of critical infrastructure protection. [24]

A. Limited Real-Time Defense Systems

While many detection sysgtems function well at detecting infections after-the-fact, working real-time defenses capable of identifying and stopping ransomware at the earliest possible stage are still elusive. Most of the modern approaches using machine learning appear to generate an excess of false alarms in critical settings such as hospitals or

industrial control systems and this hampers their operational deployment. Furthermore, many of these organizations still depend on legacy antivirus or intrusion detection systems incapable of detecting emerging or novel fileless ransomware variants in real-time. [25]

B. *Lack of RaaS Threat Attribution*

The Ransomware-as-a-Service (RaaS) model has splintered the ransomware ecosystem, making the attribution of threat actors harder than ever. Affiliates tend to recycle infrastructure, share tools, and frequently undergo rebranding which clouds forensic and law enforcement analysis. Existing literature tends to emphasize the technical aspects and overlooks creating frameworks that attribute specific ransomware attacks to certain actors or groups of affiliates, especially in the context of transnationalism. [26]

C. *Insufficient Defense Policies Against Ransomware Specific to Sectors*

Even with disproportionate focus on critical infrastructure, there remains a striking void of tailored tailored defense policies or compliance mandates for individual sectors. As an example, the energy sector with its NERC CIP might have some sector specific guidelines, but healthcare, logistics, and even water systems frameworks are out of date, if they exist at all. Research and regulatory organizations tend to overlook sensitivity of data and operational constraints specific to each sector when developing general overarching strategies to mitigate ransomware. [27]

D. *Need for Active Hacking and Threat Simulation*

Currently, most research deals with post incident response, actively injecting ransomware measures, while proactive threat pursuit remains left largely unexplored in practice. Before deploying encryption routines, advanced persistent ransomware groups perform lateral movement and reconnaissance deep into a system or network. Active simulation of attacks, red teaming, mock adversary emulation, as well as campaign-based emulation – often enumerated in the MITRE ATT&CK framework – are not practiced by many organizations. Academic literature focused on effective techniques for hunting threats in air gaps or ICS/SCADA systems is sparse.

VII. Results and Discussion.

Analysis of the aforementioned systematic review demonstrates the evolution patterns and techniques used in ransomware and defense mechanisms post-2017.The numbers reveal a dramatic shift from opportunistic ransomware to more strategic ones. What it suggests is that cybercriminals are more inclined towards high-value industries where time is expensive and real-time response is essential.

This shift is further exacerbated with the emergence of double and triple extortion models that leverage reputation, finances, and legal threats to demand payment alongside data encryption.

A. *Most Targeted Sectors.*

The most affected critical infrastructure sectors are healthcare, energy, transport, and manufacturing, which are explained in the review to be impacted by the fact that they are reliant on legacy systems, do not have real-time monitoring and require hard uptime. These hospitals were very vulnerable to attack during the COVID-19 pandemic. Additionally, the 2021 Colonial Pipeline attack demonstrated how ransomware attacks can have rippling national effects.

B. *Emerging Advanced Diagnostic Defenses*

The most impactful defenses described in the literature include the following:

- Exceptional Behavior Detection Systems: These Anomaly Detection Systems detect suspicious actions within user activity or network traffic prior to encryption.
- Anomaly Detection Systems: ZTA (Zero Trust Architecture) reduces lateral movement by implementing strict access control and continuous verification.
- AI/ML-based detection: These methods require a significant amount of tuning to minimize false alarms but prove effective at large scale data scrutiny during the recognition of breach precursors.
- Executive frameworks like MITRE ATT&CK serve for simulation and understanding of organization's preparedness level against real-world ransomware tactics.

Although these are promising defenses, there is a lack of balanced preventive delay tools that operate effectively during critical environment without heightening sensitivity in real time. [6]

Highlighting a stark reality of the defense-sophistication-threat-gap, this study notes, all infrastructure levels face continuous static struggles against evolving adversarial defenses.

Overall, the balance between emerging threats and available defenses posed the most sophisticated arms race.

VIII. Conclusion

This systematic literature review focuses on the changing ransomware methods and countermeasures implemented within the infrastructures year after year from 2017 to the present. It is certain that the world of ransomware has transformed from opportunistic, straightforward intrusions into well-organized, advanced attacks employing double extortion, triple extortion, and Ransomware-as-a-Service (RaaS). These developments have greatly increased both the

number and intensity of ransomware attacks, particularly in critical industries like healthcare, energy, transportation, and manufacturing.
The need to develop more comprehensive cybersecurity measures within critical infrastructure systems cannot be overstated. The maintenance and societal impact of fuel pipelines, public healthcare services, and various other public services being made nonoperational showcases that ransomware is far more advanced than just an irritant; it's a precision risk strategy aimed at undermining national and economic security.

Strategically, the need for future research and practice is focused on creating a combination of AI, behavioral analytics, and specific sector resilience for adaptive, real-time defenses. In parallel to technical responses aimed at neutralizing the widening disparity between what the attackers can do and what organizational defenses are set up to withstand, the need for international collaboration on attributing threats and enforcing policies against ransomware needs to become a focal point.

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AUTHOR PROFILE

JAYARATHNA K.P.G.C.M is a cybersecurity enthusiast currently pursuing a bachelor’s degree at the Sri Lanka Institute of Information Technology (SLIIT). He excels in identifying vulnerabilities and putting preventative security measures in place because he concentrates on penetration testing and ethical hacking. His passion for increasing knowledge and understanding of the subject of cybersecurity is demonstrated by his active involvement in workshops and seminars, which demonstrates his commitment to security awareness and education